EE4-51
EE9-FPN2-11

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

Corrected Copy

EEE PART IV: MEng and ACGI

POWER SYSTEM ECONOMICS

Tuesday, 14 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Power System Economics – Exam Questions 2019

Question 1

(a) Explain the significance of the economic (merit order) dispatch, describe the significance of the system marginal price and explain why it varies with time in the case of electricity. [4]

(b) Two generators are available to supply demand in three consecutive hours: $D_1 = 250\ [MW]$, $D_2 = 960\ [MW]$ and $D_3 = 340\ [MW]$. The cost of generating power $C_{i,t}(P_{i,t})$ corresponding to each of the generators i in hour t is:

$$C_{1,t}(P_{1,t}) = 660 + P_{1,t} + 0.065P_{1,t}^2\ [\frac{£}{h}]$$

$$C_{2,t}(P_{2,t}) = 470 + 15P_{2,t} + P_{2,t}^2\ [\frac{£}{h}]$$

Calculate:
- the optimal production of each generator at each hour,
- the system marginal cost at each hour,
- the total cost of operating the system, and
- the total profit for each of the generators.

Which hour is characterised by the highest system marginal cost and why?

[6]

(c) Modify the dispatch found in **(b)** that respects to the constraints related to the minimum and maximum outputs for each of the generators:
$P_1^{min} = 100\ [MW]$ and $P_1^{max} = 900\ [MW]$
$P_2^{min} = 50\ [MW]$ and $P_2^{max} = 300\ [MW]$
Calculate:
- the optimal production of each generator at each hour,
- the system marginal cost at each hour,
- the total cost of operating the system, and
- the total profit for each of the generators.

[8]

(d) The total profit of one of the generators in **(c)** is negative. Explain the reason for this as well as why and how it should be avoided in practice. [2]

Question 2

(a) Provide the key reasons for the electricity market not being perfectly competitive. [2]

(b) What are the conditions for perfectly competitive market? How can a generation company in an imperfect competitive environment influence the market price? What is the relationship between the marginal cost of production and the market price in perfectly and imperfectly competitive markets? [3]

(c) Consider a market for electrical energy that is supplied by two generating companies whose cost functions are:

$C_A = 75 * P_A \left[\frac{£}{h}\right], C_B = 62 * P_B \left[\frac{£}{h}\right]$

Where P_A and P_B are production levels of the two generating companies respectively. The inverse demand function in this market is estimated to be:

$$\pi = 250 - 2D \left[\frac{£}{MWh}\right]$$

Where π and D are the electricity price and energy demand respectively.

i. Assuming Bertrand competition model in this market, calculate the production of each of the companies, the resulting market price and profits made by both companies. [3]

ii. In order to identify the Nash Equilibria (NE) in this market under the Cournot competition model, form a table for the range of production levels for company A of 20MW, 25MW, 30MW and 35MW, and for company B of 25MW, 30MW, 35MW and 40MW, considering key parameters as presented in Table 2.1.

Table 2.1 Key parameters for assessing Nash Equilibrium

D (Demand)	Ω_A (profit A)
Ω_B (profit B)	π (Price)

Identify production levels of companies A and B that correspond to two Nash Equilibria cases in this example. [6]

(c) It is observed in (b)-ii that one of the cells of the derived Cournot table involves higher profits for both competing companies, than the ones corresponding to the NE. Explain why this cell does not express the equilibrium state of the market. [2]

(d) State the optimality conditions and evaluate the exact NE of this market, including equilibrium market price, system demand and profit of each company. [4]

Question 3

Part I: Consider a centralised electricity market that consists of four generating units, each unit produces/sells electricity by bidding at its marginal cost. Table 3.1 shows their relevant technical and cost characteristics. Figure 3.1 presents the amount of reserve that generating units 2 and 3 can provide as a function of their power output. Assume that the system demand at a particular point in time is 500MW and that 285MW reserve is required.

Table 3.1. Marginal cost, maximum output and reserve capability of generating units

Generation Units	Marginal cost (£/MWh)	Max Power output capacity (MW)	Max reserve power capacity (MW)
1	5	300	0
2	13	250	180
3	21.5	270	220
4	30.2	265	0

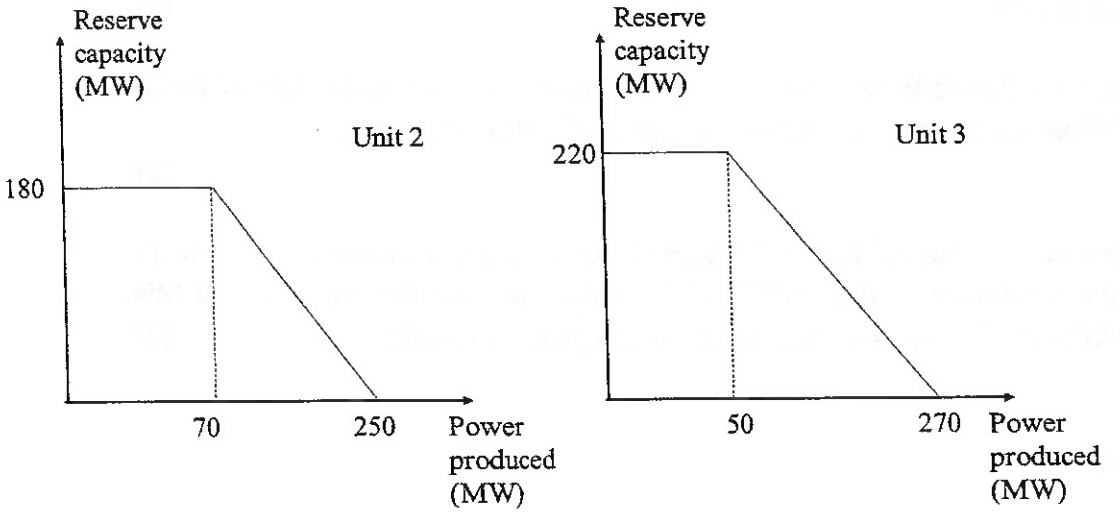


Figure 3.1. Reserve provision of generating units 2 and 3 as a function of the amount of power produced

(a) Determine the optimal power produced and reserve provided by each of the generating units. [4]

(b) Determine the electricity production price and reserve price. [3]

(c) Determine the combined profit that generation unit 2 derives from participating in energy and reserve markets. [3]

Part II: Borduria Power is considering investment in a 400-MW power plant. On average, the plant is expected to operate at a maximum capacity for 7446 hours per year and sell its output at an average price of 31 £/MWh. The parameters of the plant are presented in Table 3.2.

Table 3.2 Assumed data for the plant

Investment cost	1200 £/kW
Expected plant life	30 years
Heat rate at rated output	9800 Btu/kWh
Expected fuel cost	1.10£/MBtu

In order to assess the business case of this investment, the following questions need to be answered:

(a) Calculate the investment cost, estimated annual energy production, annual production cost, annual revenue and annual profit. Calculate the total cash flow over the expected plant life. [5]

(b) Explain how should Borduria Power set its Minimum Acceptable Rate of Return (MARR)? What are the potential risks associated with this investment? [2]

(c) Explain how the Internal Rate of Return (IRR) should be calculated and how it is related to the Net Present Value (NPV). If the IRR of net cash flow stream is 12.14%, and the MARR is 12%, can the investment in this plant be justified? [3]

Question 4

(a) Briefly explain the key benefits of a transmission network. [2]

(b) In a competitive electricity market, transmission network is often separated from energy trading. List and briefly explain key characteristics of transmission as a stand-alone business. [4]

(c) For the system shown in Figure 4.1, and the data given in Tables 4.1 and 4.2, calculate the unconstrained economic dispatch (i.e. ignoring capacity of transmission lines) as well as the resulting nodal prices and power flows in each line. [4]

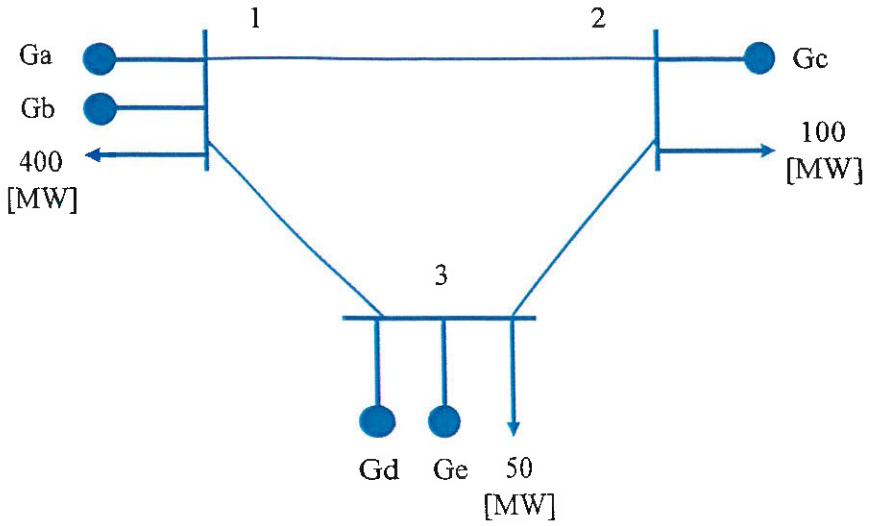


Figure 4.1: System layout.

Table 4.1. Generation data

Generator	Capacity (MW)	Marginal cost (£/MWh)
Ga	100	20
Gb	120	15
Gc	150	12
Gd	200	10
Ge	300	8

Table 4.2 Network data

Line	Per unit reactance	Capacity (MW)
1-2	0.20	250
1-3	0.25	250
2-3	0.25	250

(d) Demonstrate that one of the lines is overloaded if the capacity limits of the lines are considered and the dispatch calculated in **(c)** is implemented. Investigate if this overload can be eliminated by:
i) increasing the output of generator B and decreasing the output of generator D
ii) increasing the output of generator C and decreasing the output of generator D
Calculate the hourly cost of options i) and ii) above. Which one of the two options is preferable and why? [5]

(e) Calculate the nodal prices when the more preferable of the two options compared in **(d)** is implemented. [5]